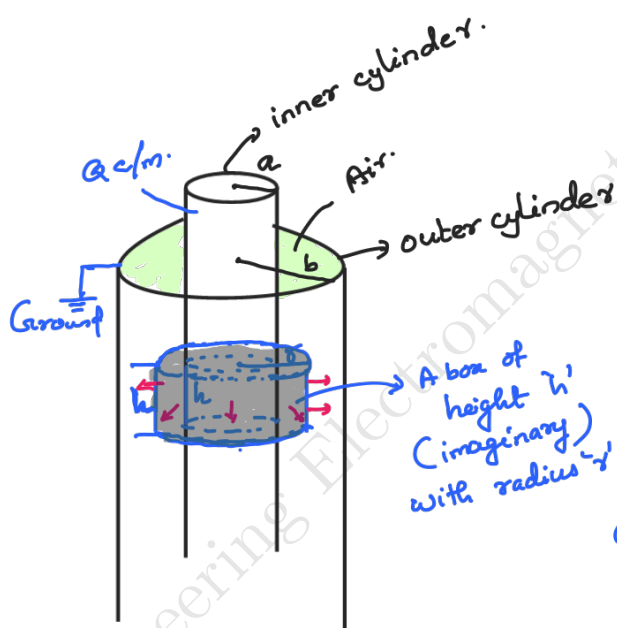


Gauss's law ;

$$\oint \vec{E} \cdot d\vec{s} = \frac{Q_{\text{enclosed}}}{\epsilon_0} \quad ; \quad \nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} .$$

Example 1

* Cylinders made of conductors .

* Region between cylinder is empty / air.

* Amount of charge put in inner cylinder $Q \text{ C/m}$.Obtain $E(\vec{r})$ & $\phi(\vec{r})$ in $a < r < b$.Solution: Consider an imaginary box of radius r , height h .

Electric field is radially outward.

Hence in the top and bottom surfaces of the box,

Electric field is tangential to the surface

$$\Rightarrow E_r \cdot d\vec{s}_{\text{upper}} = E_r \cdot d\vec{s}_{\text{lower}} = 0 .$$

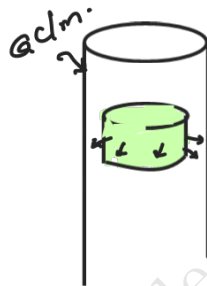
$$\Rightarrow \oint \vec{E} \cdot d\vec{s} = E_r(r) 2\pi r h = \frac{Qh}{\epsilon_0}$$

Total charge in height h : $Q \frac{h}{\pi a^2 h} = Q \frac{h}{\pi a^2}$

$$E_r(r) = \frac{Q}{2\pi r \epsilon_0}$$

$$\vec{E}(r) = \frac{Q}{2\pi \epsilon_0} \cdot \frac{1}{r} \hat{r}; \text{ for } a < r < b.$$

Now consider a box inside the inner cylinder $\Rightarrow r < a$.

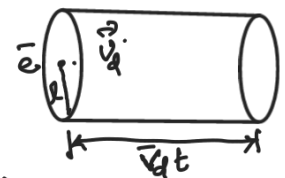
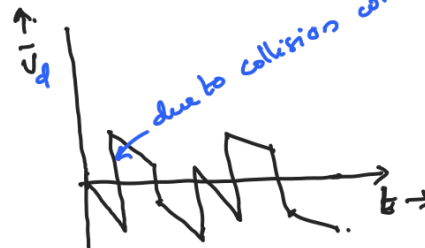


$\vec{E}_r$? is it possible?

Conductors $\rightarrow$ Free electrons.

$\vec{e}^- \rightarrow$ experience force $\vec{E} \vec{e}^-$

Velocity of $\vec{e}^-$ is;



On an average the drift velocity $\langle \vec{u}_d \rangle$ is constant in conductors.

Force experienced by $\vec{e}^-$

$$m_e \frac{d\vec{u}_d}{dt} = -\vec{E} \vec{e}^- - m_e \nu \vec{u}_d$$

Collision frequency $\propto$ density of $\vec{e}^-$

Force due to collision.

This term is an approximation. More or less true in metals & semiconductors.



But at steady state ;
meaning on an average over time $\langle v_d \rangle \approx \text{Constant}$.

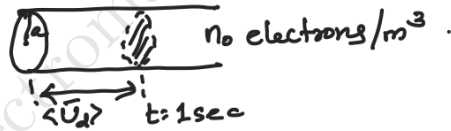
$$\Rightarrow 0 = -\langle \vec{E} \rangle e - m_e v \langle \vec{v}_d \rangle$$

Average drift velocity $\Rightarrow \boxed{\langle \vec{v}_d \rangle = \frac{-e \langle \vec{E} \rangle}{m_e v}} \quad \text{ie; } \langle \vec{v}_d \rangle \propto \text{Charge/mass.}$
 $\langle \vec{v}_d \rangle \propto \text{Applied electric field}$
 $\langle \vec{v}_d \rangle \propto \frac{1}{\text{number of } e^- \text{ present}}$

---- (1)

Current : It is simply moving electrons. (Charge crossing per unit time)

So if we have wire with n_0 electrons/ m^3



$$I = \underbrace{\pi a^2 \langle v_d \rangle n_0 e}_{\text{number of } e^- \text{ crossing the cross section } \pi a^2 \text{ in 1 sec.}} \quad \text{--- (2)}$$

$$\# \pi a^2 \langle v_d \rangle = \text{Volume.}$$

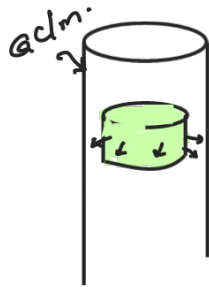
$$\# n_0 e^- \times \text{Volume} = \text{Total number of } e^- \text{ passing the cross section in 1 sec.}$$

From (1) $\langle v_d \rangle \propto E$

(2) $\Rightarrow \text{Current } I \propto \langle v_d \rangle$

$\Rightarrow \text{Current} \propto \text{applied } E.$

Let us consider the inner box inside inner cylinder.



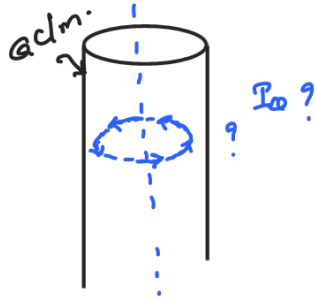
$\vec{E}_r$? is it possible?
 $I = ?$

* IF $\vec{E}_r$ present $\Rightarrow$ Current present and current flows out $\Rightarrow$ Charge or e^- keeps moving out.
Is it possible to have the continuous flow of e^- radially outward without replacing the moving e^- from inside?

Answer is $\Rightarrow$ No.

ie; $I = 0$
 $I \propto E_r \Rightarrow \boxed{E_r = 0}$ inside conducting material.

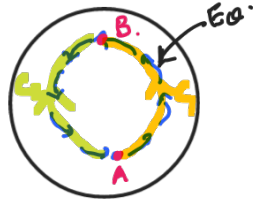
Now, can current flow in theta (θ) direction without requiring replacement of e^- ? ie Is it possible to have circulating current in electrostatic condition here?



If I_0 is present
ie; $I_0 \propto E_0$.

Now the question :
Is E_0 present inside?

Take cross section;



$\int_A^B \vec{E} \cdot d\vec{l} =$ it should be independent of path.

$$\int_{C_1} \vec{E} \cdot d\vec{l} = +ve.$$

$$\int_{C_2} \vec{E} \cdot d\vec{l} = -ve.$$

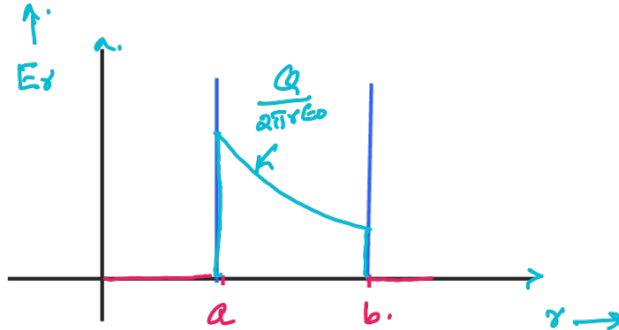
$$\Rightarrow E_0 = 0 \Rightarrow I_0 = 0.$$

* Cannot have circulating current in Electrostatics.

Precautionary Note: This is the case for Electrostatics.
Later we will introduce Faraday's law
where we can have circulating electric
current in a conductor while we
deal with magnetic field (time varying
currents).

Here in summary; Electric field is zero inside a conducting material.

So in Co-axial cable;



* Think and tell the possible reasons for $E_r = 0$ for $r > b$?

Now if we know potential at one of the cylinders:

Say $\phi(a) = V_0$.

$$\phi(r) = - \int_a^r \vec{E} \cdot d\vec{l} + \phi(a)$$

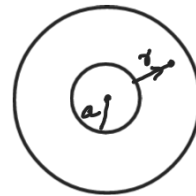
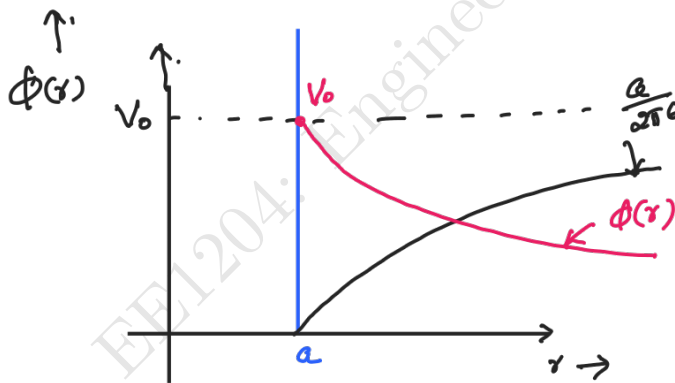
$$= - \int_a^r E_r dr + \phi(a) = - \int_a^r E_r dr + V_0$$

$$= - \frac{Q}{2\pi\epsilon_0} \int_a^r \frac{dr}{r} + V_0$$

$$= - \frac{Q}{2\pi\epsilon_0} (\ln r - \ln a) + V_0$$

$$\phi(r) = V_0 - \frac{Q}{2\pi\epsilon_0} \ln\left(\frac{r}{a}\right)$$

Slowly varying potential w.r.t 'r'.



In summary;

$$\oint \vec{E} \cdot d\vec{s} = \frac{Q_{\text{enc}}}{\epsilon_0} \Rightarrow \nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}, \dots (1)$$

Also, $\vec{E} = -\nabla\phi \dots (2)$

Given $\phi \rightarrow \vec{E} = -\nabla\phi \rightarrow \nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \leftarrow \text{scalar.}$
 scalar $\uparrow$ vector

$$\phi \rightarrow \vec{E} \rightarrow \rho$$

Similarly $\rho \rightarrow \vec{E} \rightarrow \phi$

But can we move back and forth between ϕ & ρ ?

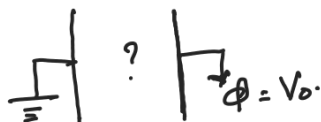
We know $\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$

But $\vec{E} = -\nabla\phi \Rightarrow \nabla \cdot (-\nabla\phi) = \frac{\rho}{\epsilon_0}$

$$\Rightarrow \nabla \cdot \nabla\phi = -\frac{\rho}{\epsilon_0}$$

$$\boxed{\nabla^2\phi = -\frac{\rho}{\epsilon_0}} \leftarrow \text{Poisson equation.}$$

This Eqn $\Rightarrow$ Very important to study electric field distribution in capacitors!



We cannot apply Coulomb's law here but use Poisson's Equation!

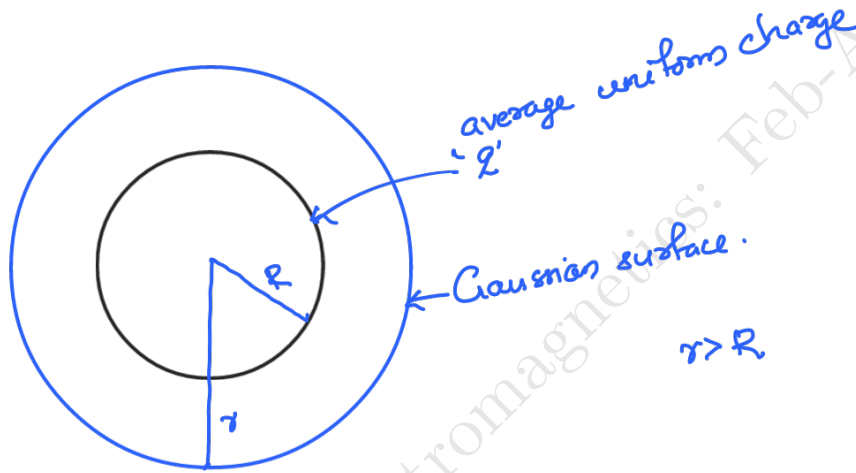


Applications of Gauss's law

Example: 2.

Find the field outside a uniformly charged solid sphere of radius R and total charge Q .

Solution:



$$\oint_S \vec{E} \cdot d\vec{s} = \frac{Q_{enc}}{\epsilon_0}$$

$$E_r \int r^2 \sin\theta d\theta d\phi = E_r 4\pi r^2 = \frac{Q_{enc}}{\epsilon_0} \quad ; \quad Q_{enc} = Q$$

$$|E| = \frac{Q}{4\pi\epsilon_0 r^2}$$

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r}$$

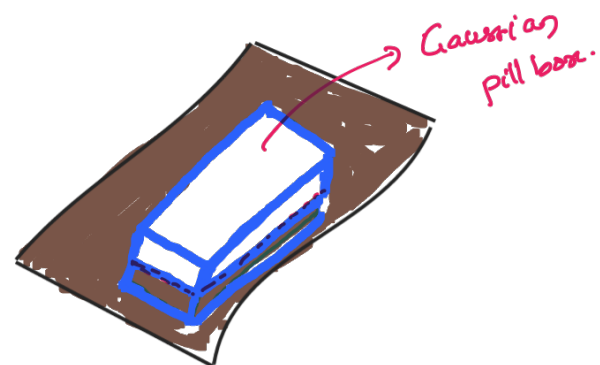
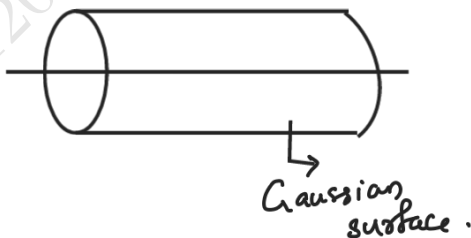
* Field outside the sphere = field if all the charge had been concentrated at the center.

* Gauss's law is always true; but is it always useful?
 - If here ρ was not uniformly distributed charge or not spherically symmetric
 $\Rightarrow$ Without symmetry (in example 2, if we choose some other shape than sphere as gaussian surface,) we cannot take $|\vec{E}|$ outside surface integral, as it might not be constant over the surface.

Symmetry is crucial for applying Gauss's law.

Only three kinds of symmetry work.

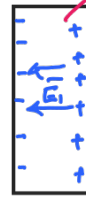
1. Spherical symmetry : Gaussian surface $\rightarrow$ Concentric sphere.
2. Cylindrical symmetry : Gaussian surface $\rightarrow$ co-axial cylinder.
3. Plane symmetry : Gaussian "pill box" that saddles the surface.
(Cartesian-coordinates)



Electrostatic properties of ideal conductor.(i) $\vec{E} = 0$, inside a conductor.

Free electrons moves in the opposite direction, keeping +vely charged Neutron. Basically the charges piles up near the surface inducing electric field $\vec{E}_i$ which exactly cancel out $\vec{E}_0$.

* Or the charges moves till the field $\vec{E}_i$ due to induced charges cancel the applied field.



$\vec{E}_0 \leftarrow$ external electric field.

(ii) $\rho = 0$ inside a conductor: From Gauss's law $\nabla \cdot \vec{E} = \rho / \epsilon_0 \Rightarrow \vec{E} = 0 \Rightarrow \rho = 0$.
* The net charge density in the interior is zero.

(iii) Any net charge reside on the surface.

(iv) A conductor is an equipotential: For any 2 points "a", "b" within (or at the surface of) the conductor, $\phi(b) - \phi(a) = - \int_a^b \vec{E} \cdot d\vec{l} = 0$
 $\Rightarrow \phi(b) = \phi(a)$

(v) $\vec{E}$ is perpendicular to the surface, just outside the conductor.
(understand this concept w.r.t the point (i))